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Zone-Aware Pneumonia Classification Using Automated Lung Region Detection and Multi-Branch Feature Learning

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Abstract
Pneumonia is a major lung infection and remains one of the leading causes of death, especially among children under five and older adults. Early and accurate diagnosis often depends on chest X-rays, yet manual interpretation is challenging due to overlapping structures and low-contrast opacities. In this context, computer-based techniques have been proposed to assist radiologists in detecting pneumonia symptoms. Most existing techniques rely on deep learning models that extract features from the entire image to make classification decisions. While these approaches achieve high accuracy, they lack explainability, offering no clear information on where features are localized or how decisions are made. In this paper, we address two main limitations: (1) the localization of pneumonia-related features, and (2) the interpretability of deep features used for decision-making. This research presents a robust framework that integrates lung zone detection with multi-branch feature learning to classify pneumonia types and predict zone-specific findings. The proposed system consists of three phases. First, lung zone localization is performed to automatically focus on disease-relevant regions. Second, intra-zone CoT reasoning is introduced for zone-specific feature extraction, where lung zones detected by YOLOv8 are processed by ResNet-18 to capture local features. Third, attention-based inter-zone CoT fusion and classification is applied to predict pneumonia type and zone-level findings such as opacities and ground-glass patterns, aligning with radiology reporting standards. The proposed framework is evaluated on the Chest X-Ray Images (Pneumonia) dataset. It achieved an average accuracy of 94.4% and demonstrated notable improvements in detecting early-stage pneumonia. These results highlight the potential of the model as a decision-support tool for radiologists, enabling accurate diagnosis and standardized reporting in clinical practice.

Keywords Pneumonia detection · Deep learning · YOLOv8 · Zone detection · Lung disease · Chain-of-Thought (CoT)

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1 Introduction

Pneumonia is a lung infection caused by bacteria and viruses, affecting people of all ages, including newborns. It is considered one of the most serious respiratory diseases worldwide [1]. Early detection with timely treatment is the major aspect to prevent complications. Pneumonia also remains a leading cause of death globally, especially among children under five and adults [2]. According to studies, the United States has one million cases of pneumonia annually, which leads to 50,000 deaths each year [3]. Generally, the risk is higher in newborns than in other age groups, and in such cases, delay in diagnosis and inappropriate treatment can lead to worse infection [4]. Various symptoms of this disease include cough, fever, and difficulty breathing. Bacterial pneumonia is usually treated with antibiotics, while viral pneumonia may require supportive care [5]. However, symptoms and treatment are not sufficient for an accurate diagnosis. Pneumonia is traditionally detected through physical examination, patient history, laboratory tests, and chest X-ray analysis by experienced specialists [6]. Sometimes, pneumonia appears as abnormal in a patient's chest radiographs, due to overlapping symptoms with various other lung conditions, including pulmonary edema, hemoptysis, lung collapse, and neoplasm, which occur from post-surgical alterations or radiation pneumonitis and require further diagnostic techniques [7].

At initial level, treatment begins with evaluation of early clinical symptom such as cough, fever, chest pain, and analyzing lung sound with a stethoscope followed by radiological imaging to identify and measure severity of infection [8]. Therefore, accurate diagnosis often requires repeated Chest X-rays (CXRs), clinical symptoms and examine the detailed patient's medical history [9]. Chest x-rays is a primary diagnostic tool but variation in radiologist expertise and judgment can lead to error, delay in diagnosis and also make accurate detection difficult [10]. These challenges highlight the need for added computer systems based on artificial intelligence (AI) that can accurately detect pneumonia. Moreover, in recent years, the rapid growth of medical imaging data has driven the need of more robust automated diagnostic tool [11]. For the development of such diagnostic tools imaging data is integrated with advanced AI approaches that emerged as a promising field in pneumonia diagnosis and classification, and in other broader medical practices. These AI-based approaches, such as synthetic image augmentation, ensemble learning, achieve higher efficiency in healthcare, fast radiograph analysis, minimum human errors [12]. Among these techniques, deep learning algorithms have demonstrated exceptional outcomes in categorizing pneumonia from CXRs and CT scans that address the limitation of traditional methods [13]. Following these advances, recent evaluations highlight the development of medical imaging-based AI systems integrated with advanced algorithms to reliably recognize pneumonia-related irregularities in chest images [14]. These hybrid models not only beneficial in enhancing user trust but also accelerating safer deployment especially in resource-limited settings [15]. AI systems integrate various techniques, including machine learning, deep learning, and natural language processing, to mimic human decision-making and improve diagnostic accuracy. In pneumonia detection, these hybrid models can automatically identify abnormalities in lung regions, classify the type of infection, and even provide reasoning based on radiology reports. These intelligent systems enhance clinical workflows by offering faster assessments, reducing human error, and supporting radiologists in handling large volumes of patient data efficiently. As AI continues to evolve, its role in early detection and personalized treatment planning for pneumonia is expected to grow significantly. Despite these high accuracies, AI-based deep learning model are often criticized for their "black-box" nature, raising questions regarding trust and transparency in clinical usage [16]. Beyond this lack of interpretability, the shortage of radiologists, high patient volume, and variability in radiographs have further motivated the adoption of Explainable AI (XAI). XAI enables clinicians to read and comprehend the model outcome while assisting in the analysis of chest X-rays, other medical images. These systems provide transparency in decision-making, and detect early pathological signs that may be overlooked using deep models [17]. Eventually, they improve diagnostic accuracy, enhance clinical confidence, and give access to quality healthcare. Further techniques such as Gradient weighted Class Activation Mapping (Grad-CAM) [18]and Local Interpretable Model Agnostic Explanation (LIME) [19] help in visualizing the affected region of an image that influences the model in the diagnosis process. Integrating XAI with deep learning hybrid models allows doctors to better understand how a model detects, this interpretability plays an important role in decision-making that needs to be transparent and explainable. These XAI models provide both high diagnostic accuracy and visual interpretability, which helps to gain trust of medical practitioners and enhances diagnostic confidence [20]. The importance of such systems became clear during pandemics or other critical health emergencies when AI tools were needed to quickly distinguish between various types of pneumonia, since their signs on radiological images can be very similar. To address this, a zone-aware pneumonia classification approach is introduced, and the main contributions are listed below:

- An automated zone-aware pneumonia classification system is developed that incorporates YOLOv8 to locate all five clinically relevant zones of parenchyma for region specifical analysis.
- A modified ResNet-18 is adapted to extract features from all five zones of lung parenchyma obtained from YOLOv8 and these features are passed to the proposed dual-chain CoT for attention guided classification. The goal of this step is to capture internal dependencies and subtle patterns.
- A multi-branch CoT-based fusion module is developed that applies attention to weight each zone. This module integrates inter zone details with global context that enhance interpretation by creating zone-specific Grad-CAM heatmaps.
- A modified ResNet-18 is adapted to extract features from all five zones of lung parenchyma detected by YOLOv8. These features are passed through proposed dual Chain-of-Thought (CoT) reasoning to capture intra- and inter-zone dependencies and patterns for accurate pneumonia classification.
- A multi-branch CoT-based fusion module is developed that applies zone-wise attention to integrate local and global features. This module enhances interpretability by creating zone-specific Grad-CAM heatmaps that highlight clinically relevant regions.

This research is organized as follows. Section II addresses the state-of-the-art methods for pneumonia classification. Section III presents a proposed methodology. Section IV presents the experimental evaluation. Section V concludes with summary and discussion.

2 Literature Review

Pneumonia is a potentially life-threatening respiratory infection affecting individuals of all ages, particularly young children and the elderly [21]. With the growing worldwide cases of pneumonia, traditional diagnostic methods, interpreted by radiologists, often face limitations [22]. These challenges are especially significant in low-resource settings where access to advance clinical tools is limited. Therefore, recent research has increasingly focused on leveraging artificial intelligence (AI) and deep learning techniques to enhance diagnostic precision and support early detection [23]. This literature review aims to explore and critically analyze the advancements in pneumonia classification and diagnosis, highlighting the role of AI-based solutions and their effectiveness across various studies.

Deep learning has emerged as a transformative approach for automated detection of pneumonia, particularly through convolutional neural networks (CNNs), that outperform in analyzing medical images such as chest X-rays and CT scans. These models learn hierarchical features directly from raw pixel data, enabling them to identify complex visual patterns associated with pneumonia. To address this, several studies have worked to improve CNN-based approaches that can effectively extract the main features and detect pneumonia from X-ray images. Al-Azzawi et al. [24] developed a deep CNN that applies custom network layers and different optimized trained techniques to detect pneumonia in a chest X-ray and obtained robust and accurate results. Multiple experiments are conducted that demonstrate the effectiveness of the model on large datasets. This technique serves as a reliable tool for medical professionals, enabling fast and accurate detection in pneumonia patients. Shilpa et al. [25] also introduced a CNN-based solution design to make pneumonia recognition from chest X-rays more efficient. By addressing the limitation of manual diagnosis, which can be inaccurate and take a lot of time, an AI system based on CNNs is introduced. This approach relies on sophisticated preprocessing, image enhancement, and optimized model training to improve results. The study clearly demonstrates that their AI can detect pneumonia reliably and consistently across diverse health data, contributing significantly to faster and more uniform pneumonia screening in healthcare settings. Similarly, Beri et al. [26] employed convolutional neural networks (CNNs) within an AI-driven framework for detecting pneumonia on chest X-rays. Their work aimed to overcome the problems of subjectivity and low consistency in results after manual interpretation. By training CNN specifically on chest radiographs, the proposed solution achieves better classification of pneumonia. The analysis evaluated multiple optimization approaches and loss functions, demonstrating that well-configured CNN models can outperform convolutional diagnostic methods. Additionally, this study introduces an effective detection approach that advances AI in radiology for pneumonia diagnosis. Another CNN-based AI system was developed by Almohab et al. [27] to automatically detect pneumonia from chest X-rays. This study examined how convolutional neural networks can be tuned to decrease errors during diagnosis. Due to the size-limited CNN, this system is reliable for deployment in severely resource-limited areas. The model's performance is measured by metrics such as accuracy, precision, and F1-score, and shows strong results on widely used benchmark datasets. Their work is efficient and provides a practical method for early pneumonia detection in clinical settings with limited computing resources. Nettur et al. [28] proposed a weighted average combination of different CNNs for the detection of pneumonia in chest X-rays. This approach addresses the limitation of robustness in a single model by integrating multiple CNN architectures. Their technique improves detection accuracy while reducing the computation requirement. CNN models provide better performance but are often limited to pneumonia detection.

To overcome this gap, researchers expanded their work into multimodal AI systems that can handle pneumonia and other lung-related diseases. Early research focused on single disease detection such as Howroyd et al. [29] presented AI-based diagnostic technique to analyze ventilator-associated pneumonia (VAP), a common critical disease. In this technique, clinical, microbial, and immune tests are integrated to identify several groups of VAP and provide targeted strategies to diagnose them. The results suggest that using AI for diagnosis should take into account the wide variety of biological factors, so that new systems can suggest personal pneumonia management strategies. After single disease detection research targeted different diagnoses, for example, Ippolito et al. [30] integrating AI with chest X-rays helped with making a differential diagnosis of lung pneumonia in emergency cases, as reported in Diseases. The model they developed could learn to identify pneumonia apart from similar lung diseases like heart failure or pulmonary edema. The research stresses the importance of real-time use in emergency departments, since accurate triage needs to happen very quickly. The tool shows it can reliably assist clinicians in stressful settings, allowing them to provide care quickly and consistently. Recent research has shifted toward multi-disease frameworks, such as OView-AI is a system introduced by Oh et al. [31] that works in cycles to distinguish pneumonia, pneumothorax, tuberculosis, and lung cancer from chest X-rays, according to the paper in Diagnostics. The approach separates the image into superpixels so that analysis is conducted locally and improves predictions with a hierarchy of classifiers. By paying attention to affected regions, this procedure can better explain and refine the results. OView-AI is better than regular CNNs in detecting various diseases in images and gives valuable help in making comprehensive radiography diagnoses in places where resources are scarce. AI-based multimodal techniques are effective for single-disease cases like VAP, but there are many novel techniques that target multiple lung conditions in one analysis such as Kabir et al. [32] present a system powered by AI that can identify chest radiographs showing COVID-19, pneumonia, and tuberculosis using knowledge distillation. The study, which appears in Heliyon, seeks to simplify deep learning models to preserve their level of excellence. Since the framework guides a small student model using a more complicated teacher model, both tasks are handled more efficiently, and memory usage is lowered. Because it achieves strong classification accuracy on all three diseases, the model is suitable for low-resource diagnostics. Another AI-driven diagnostic framework to recognize pneumonia, tuberculosis, and COVID-19 from chest radiographs is presented by Yadav et al. [33]. To identify and categorize lung patterns linked to each disease, the authors put together a variety of deep learning models. Pre-trained CNNs are included in the framework for extracting features, while combining decisions to lower the number of false positives. The model has been worked on various datasets and reaches solid results on all three diseases. This device can identify several diseases at once, which is very useful for doctors in pandemic or limited health care situations. Furthermore, several studies have explored the improvement of multimodal learning to predict the COVID-19 such as K. Karthik et al. [34] integrate x-rays images with diagnostic reports that provide an evidence based diagnosis. They designed a dense neural network based convolutional attention model that is trained on disease-specific parameters by jointly important visual features with information that is embedded within unstructured clinical textual reports. They designed a comprehensive decision support system called as "CADNN" presented in [35] by employing ensemble deep learning models which is deployed on web portal where input images are received preprocessed, validate from experts and then utilized for modal training The experimental results achieved through this manually created multimodal based dataset proved that patient data and visual symptoms play an equal and important role in enhanced prediction.

Despite robust multimodal approaches, they often act black box system, lack clarity about how the prediction is made, which limits their interpretability for clinicians. AI systems can handle many lung diseases and achieve effective results, but high accuracy alone is not enough. Clinicians need a reason to understand model decisions in pneumonia diagnosis. For this purpose, many Explainable AI (XAI) models are developed as a study by Radocˇaj et al. in [36] that address how deep learning models used in Pediatric Pneumonia Diagnosis must be easy to understand. Clinicians are less likely to trust AI-based systems when information is not presented clearly. In order to solve this, they suggest adding a multi-phase feature learning process with activation pattern visualization, so that the model can mark important disease areas on chest X-rays. This new, understandable model delivers accurate results and helps explain them, which makes it way better for healthcare workers to understand and accept AI. Furthermore, Priyanka et al. [37] propose PediaPulmoDx, an open-source model that uses explainable AI to help interpret pediatric chest X-rays with DenseNet121. Identifying pediatric pneumonia is difficult because the changes seen on chest radiology are not always clear, and many AI models are not transparent. They join up-to-date data cleaning approaches with explainability methods to achieve both better classification results and greater transparency. Evidently, this reduces errors, explains predictions better, and improves the trust of frontline staff in using AI for diagnostic support. Gupta et al. [38] work towards increasing the rate of detecting pneumonia on chest X-ray images using modern artificial intelligence tools. This study confronts the issue of low accuracy in detection because images can vary, and subtle traits of disease may be missed. Using advanced ways to prepare the data and special deep learning models, the authors boost how well the model identifies signs of pneumonia. Better detection results and strong performance in multiple test cases suggest that this technique may improve early treatment and patient outcomes. A study based on transformer presented by Murat Ucan et al. [39] that automatically generate medical reports. To generate these reports a system is developed by integrating swin transformer with generative pre-training transformer (GPT) that combined visual features and clinically consistent textual features. This integration enhances radiologist confidence in accurate decision making. Ghuse et al. [40] provide a detailed study of AI approaches for detecting pneumonia, highlighting how fast AI is developing in the medical field. The study examines how AI is used in healthcare by looking at the difficulties involved with recognizing pneumonia, particularly related to accuracy and ease of interpretation in conventional AI strategies. Authors cover a range of AI models, including convolutional neural networks, various transfer learning methods, and hybrid approaches. They introduce recent advances that enhance the correctness of diagnoses and their importance to patients. It finds that there are still research gaps and points out that AI and multimodal models, especially explainable ones, can improve accuracy and use of these systems among doctors.

Recent studies show that AI can improve pneumonia diagnosis across many settings and patient groups. CNNs, multimodal models, and explainable AI each address different clinical needs. These systems can boost accuracy, speed, and trust when applied with proper validation. These strengths guide our model design. It integrates CNN feature extraction, multibranch inputs, and explainable AI to meet clinical needs.

3 Proposed Methodology

This section presents an automated lung region detection system integrated with a dual Chain-of-Thought (CoT) mechanism to classify pneumonia types as presented in Fig. 1. It has three main phases: first, the system localizes lung zones and identifies the corresponding regions of interest (ROIs). Second, intra-zone CoT is applied with multi-branch features extraction, which captures diverse patterns (such as texture, opacities, and edges) from each lung zone. During the third Phase, inter-zone CoT is applied to concatenate and integrate extracted features before passing through the classifier. This model focused on lung-zone regions analysis instead of entire image processing. Based on this localized analysis, it predicts the pneumonia type and associated zone-specific findings such as opacity and ground glass presence, which differentiates it from traditional deep learning algorithms. The detail of each step is described in sections below.

3.1 Dataset

For this research, a publicly available Chest X-Ray Images (Pneumonia) dataset [41] was utilized, comprising a total of 5,863 chest X-ray images. These images were collected from pediatric patients in retrospective cohorts aged one to five years. Out of the 5863 images, 1590 are normal, and 4273 are bacterial and viral pneumonia. The images are organized into three main folders: train, test, and validation. The training folder contains 5213 images, with 26% normal and 74% pneumonia. The test set includes 634 images, with 62% pneumonia and 38% normal. The validation folder has 16 images, evenly split with 50% for each class. Initial pre-processing steps, including Gaussian-based noise reduction and removal of blurry images, were applied to all X-ray images. Additionally, the dataset was reviewed by two expert physicians to ensure accurate training of the AI system.

3.2 Lung Zone Localization

This step involves automatic detection of lung zones from a chest X-ray image using YOLOv8 [42]. It is effective for detecting small and overlapping objects due to its additional detection head at a finer feature scale (P2). This additional layer improves the re-weighting of spatial features and enables the network to focus on overlapping patterns of lung zones that need precise localization. This improves its ability to capture fine details in localized regions [43]. Moreover, YOLOv8 can detect multiple predefined lung zones in a single forward pass. This makes the system efficient and suitable for real-time use in clinical practice. It also handles variation in patient anatomy,

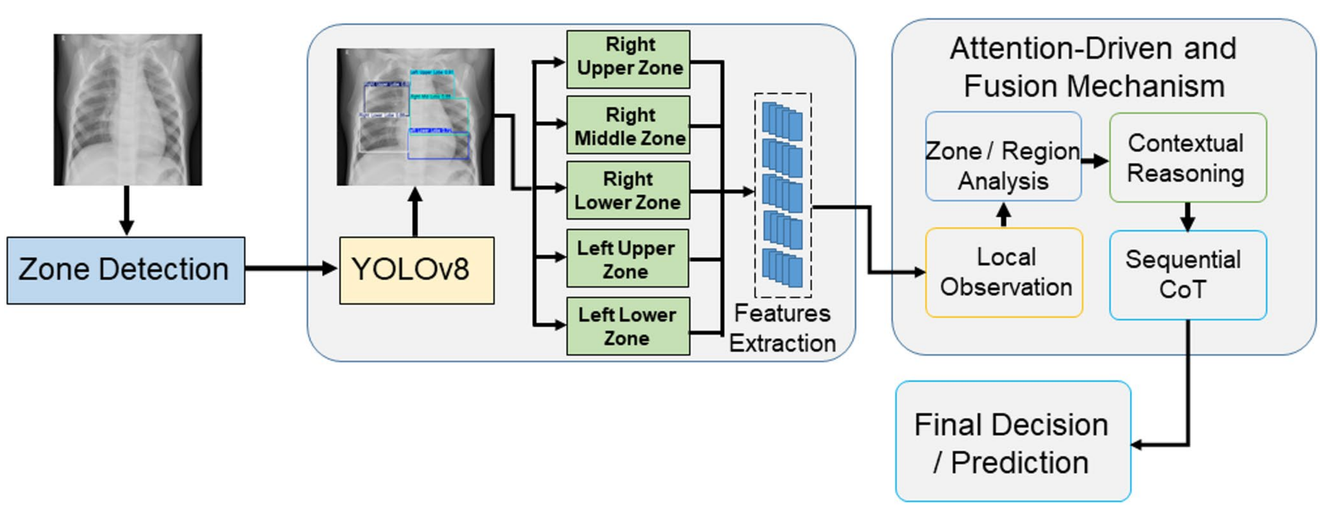

Fig. 1 Pipeline of proposed methodology for zone-aware pneumonia classification and detection

image scale, and acquisition settings without major loss in accuracy. These strengths make YOLOv8 better suited for lung zone detection than standard CNN models that rely on global features or traditional segmentation-based approaches [44].

In our formulation, this detection process is expressed as: Given as an input image $I \in \mathbb{R}^{H\times \mathrm{w}}$ is provided to the model, where H and W represent the height and width of the image, respectively. The goal is to detect predefined lung zones using a region detection model. In this research work lungs zones are not predefined or marked randomly in chest X-ray images. Anatomical criteria defined according to radiological conventions that reported in literature [45, 46] are followed, where lung field are partitioned into meaningful zones such as right upper zone, left upper zone, left lower zone, right lower zone, and middle zone which are represented as $Z = \{z_1, z_2, z_3, z_4, z_5\}$. Based on these predefined zones bounding were manually annotated on subset of training images followed by same anatomical criteria that serve as ground truth label in the detection of YOLOv8. Furthermore, each zone z_i is identified by a pixel-based bounding box which has four different coordinates $[x^i_{min}, x^i_{max}, y^i_{min}, y^i_{max}]$ where (x^i_{min}, y^i_{min}) represent the top-left and (x^i_{max}, y^i_{max}) represent the bottom right corners of i^{th} detection zone. Moreover, these bounding box coordinates are normalized internally with respect to image width, its height. The zone detected process is defined as a function:

$$Z = f_{YOLO}(I;\theta) \tag{1}$$

Where, f_{YOLO} denotes the YOLOv8 detection model trained by learnable parameters θ that represent the complete set encompassing all trainable parameters backbone, neck, feature aggregation and detection layers. Moreover, the optimization of these learnable parameters θ are performed using Admin optimizer, 0.001 (cosine decay) initial learning rate, 0.5 IoU and confidence threshold and with multi-part detection loss. Additionally, YOLOv8 uses a series of convolutional layers to predict the bounding box coordinates and associated class probabilities for each lung zone z_i from the input image I. The model is optimized using a multi-part loss function:

$$\mathcal{L}_{total} = \lambda_{box} \cdot \mathcal{L}_{box} + \lambda_{cls} \cdot \mathcal{L}_{cls} + \lambda_{obj} \cdot \mathcal{L}_{obj} \tag{2}$$

Consider an training instance example, where, $\mathcal{L}_{box}$ is a bounding box with regression loss, $\mathcal{L}_{cls}$ is the classification loss for the zone's labels and $\mathcal{L}_{obj}$ is the object score loss that contribute with weighting coefficients $?_{box},\ ?_{cls},\ and\ ?_{obj}$ to compute the total loss components. This process.

During training, for each sample, the ground truth zones $Z^{(n)}$ are annotated manually. Each training set consists of a pair $\left\{\left(I^{(n)}, Z^{(n)}\right)\right\}_{n=1}^{N}$, where, each zone z_i is represented by N is the number of training images. At inference time, given a new input image I, YOLOv8 outputs the predicted set of lung zone bounding boxes as:

$$\widehat{Z} = f_{YOLO}\left(I\right) \tag{3}$$

Where each $z_i \in \widehat{Z}$ is then cropped or masked from the original image for further feature extraction in later steps. Bounding boxes with confidence scores above a defined threshold are retained and used to crop or mask the corresponding regions from the original image. The remaining (overlapped) boxes that are below the defined region are eliminated by non-maximum suppression (NMS).

YOLOv8 offers computational efficiency and accurate localization of the lung zone. The resulting bounding box coordinates generated by YOLOv8 remove irrelevant background, highlight clinically relevant areas, and provide standardized input regions for the next Step processing. This step establishes a foundation for multi-zone feature learning.

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3.3 Multi Zone Feature Extraction

After zone detection, the multi-branch feature extraction method is extended with intra-zone CoT reasoning to refine the representation of each lung zone. Instead of extracting the feature of the entire image, this step extracts zone-specific features using ResNet-18 from every cropped lung region. Feature maps of each lung zone are processed independently in a sequence to capture fine details. This step-wise reasoning refines the feature representation and reduces information loss. In addition, it captures spatial patterns such as faint opacities, textures, and irregular densities that may be overlooked with global feature pooling. This ensures that extracted features are more clinically relevant and enhance spatial interpretability for both pneumonia classification and report attribute generation.

To process this zone-wise feature extraction, each lung zone is localized using the bounding boxes obtained from step 1. These bounding boxes $Z = \{z_1, z_2, z_3, z_4, z_5\}$ defines a sub-region of the original chest X-ray image $I \in \mathbb{R}^{H\times \mathrm{w}}$. For each zone z_i the corresponding cropped image region is:

$$I_{z_i} = Crop\left(I, z_i\right), \qquad I_{z_i} \in \mathbb{R}^{H_i\times W_i} \tag{4}$$

where $I_{z_i} \in \mathbb{R}^{H_i\times W_i}$ refers to the cropped images of i^{th} lung zone with $H_i \times W_i$ is the zone's height and width. Each cropped zone image is passed through a CNN to extract spatial features. Then these cropped images are passed to ResNet-18, where the first three pre-trained residual blocks are retained, while the fourth block and final FC layer are trained with zone-specific labels as:

$$f_{ResNet18_i} = [Block_1, Block_2, Block_3]_{pre-ytained} + [Block_4, FC]_{custom} \tag{5}$$

where $Block_1, Block_2, Block_3$ utilize frozen weights from pre-training ResNet-18 to retain low and mid-level feature representation. $Block_4$ and the fully connected (FC) layers are initialized and trained specifically with lung zone-specific labels, enabling the model to enhance its computational efficiency. Each cropped zone is passed through this network to obtain a feature map, and that feature extraction function is defined as:

$$F_{z_i} = f_{ResNet18_i}\left(I_{z_i}, p_i\right), \qquad F_{z_i} \in \mathbb{R}^{C\times H'\times W'} \tag{6}$$

where p_i represents the trainable parameters zone *i*. Where $F_{z_i}$is the output feature map and $\mathbb{R}^{C\times H'\times W'}$ is the shape of the feature map with C number of channels and $H' \times W'$ spatial dimensions across the zone. Compressing the resulting feature map F_{z_i} directly to pooling would discard information, an intra-zone CoT reasoning is applied that divides the feature map into smaller sub-features $\left\{S_i^{(1)}, S_i^{(2)}, \ldots ..,S_i^{(m)}\right\}$. These sub-features are processed step by step to retain subtle details such as:

$$E_i^{(1)} = g\left(S_i^{(1)}, S_i^{(2)}\right), \; E_i^{(2)} = g\left(E_i^{(1)}, S_i^{(3)}\right), \ldots, \; E_i^{(m)} = g\left(E_i^{(m-1)}, S_i^{(m)}\right) \qquad F_{z_i}^{CoT} = E_i^{(m)} \tag{7}$$

Where $F_{z_i}^{CoT}$ is defined as the final CoT-refined features after all reasoning steps. Here, $S_i^{(j)}$ are sub-features of the zone i with $g\left(.\right)$ is a learnable transformation and $E_i^{(k)}$ are the reasoning states.

Now, $F_{z_i}^{CoT}$ processed through global average pooling to compress dimensionality, followed by the custom FC layer that evaluates output probabilities for zone-specific attributes such as opacity presence, consolidation presence, and ground glass presence. The zone-wise classification loss is defined as:

$$\mathcal{L}_{zone} = -\frac{1}{N}\sum {}_{n=1}^{N}\sum {}_{i=1}^{5} CE(\mathcal{Y}_{z_i}^{(n)}\left(\hat{\mathcal{Y}}_{z_i}^{(n)}\right) \tag{8}$$

Where is N is the number of training samples, *CE* is the class entropy here $\mathcal{Y}_{z_i}^{(n)}$ represent the ground truth label for the zone z_i in sample n, and $\hat{\mathcal{Y}}_{z_i}^{(n)}$ is the predicted label.

Freezing the early convolutional blocks allows the model to extract and retain general low-level visual features (e.g., edges, textures) while the fourth residual block and FC layer focus on training lung zone-specific features. This not only reduces the number of trainable parameters but also improves stability on medical datasets. To enhance this representation, an intra-zone CoT reasoning step is applied across zones to capture clinically relevant spatial features and zone-level information that may be overlooked by global pooling. This approach also provides step-by-step reasoning, allowing radiologists to examine features of each zone and compare across zones. The final aggregated feature representation from all five zones serves as input for pneumonia classification.

3.4 Multi-Branch Feature Fusion

After intra-zone CoT, where zone-level features are refined. The next step is to integrate information from all lung zones. In this step, attention weighting is applied to highlight the most important zones and reduce the impact of the irrelevant ones. These weighted features are fed to the inter-zone CoT, which sequentially fuses zone-level features. This inter-CoT also helps to capture cross-zone features like bilateral opacities, multi-zone involvement, and is often missed when zones are analyzed and fused zones independently. Finally, the reasoning features are integrated with the original zone embeddings through concatenation to keep local and global details. This step provides a unified feature representation for accurate pneumonia classification and report generation that improves diagnosis.

In this step, CoT- refined zone-specific features $F_{z_i}^{CoT}$ extracted from step 2 are fused into a unified vector for final prediction. Each cropped lung zone I_{z_i} is processed through a dedicated CNN branch that generates a zone-level feature vector output as:

$$F_{z_i}^{CoT} \in \mathbb{R}^d, \qquad i \in \{1,2,3,4,5\} \tag{9}$$

where d is the feature vector dimension, and $i \in \{1,2,3,4,5\}$ corresponds to the five lung zones: right upper zone, right mid zone, right lower zone, left upper zone, left lower zone. Since not all zones of lung parenchyma contribute equally to diagnosis.

To overcome this attention mechanism is applied in which weights are learned adaptively using a SoftMax function such as:

$$\alpha_i = \frac{\exp\left(\omega^{?} F_{z_i}^{CoT}\right)}{\sum_{j=1}^{5}\exp\left(\omega^{?} F_{z_j}^{CoT}\right)}, \qquad \widetilde{F}_{z_i} = \alpha_i F_{z_i}^{CoT} \tag{10}$$

Where α_i is attention weight, which is greater if the zone is important and less for less important zones, and ω is a learnable weight. After reweighting, the features are fused step by step using the CoT mechanism instead of integrating all zones. CoT integrates features one at a time in sequence as:

$$G_1 = g(\widetilde{F}_{z_1}, \widetilde{F}_{z_2}), G_2 = g(G_1, \widetilde{F}_{z_3}), G_3 = g(G_2, \widetilde{F}_{z_4}), G_4 = g(G_3, \widetilde{F}_{z_5}) \tag{11}$$

Here, $g(\cdot)$is a learnable fusion block, G_k is an intermediate reasoning state that stores collect information of zones after each step. This step-by-step fusion where G_1 fuse the first two zones $\left(\widetilde{F}_{z_1}, \widetilde{F}_{z_2}\right)$, G_2 fuse G_1 with the third zone $\widetilde{F}_{z_3}$. Similarly, G_3 fuse G_2 with the fourth zone $\widetilde{F}_{z_4}$and G_4 fuse G_3 with the fifth zone $\widetilde{F}_{z_5}$(final fused features of all five zones.

After zone-level integration final output reasoning G_4 is concatenated with each individual zone embedding via simple concatenation as:

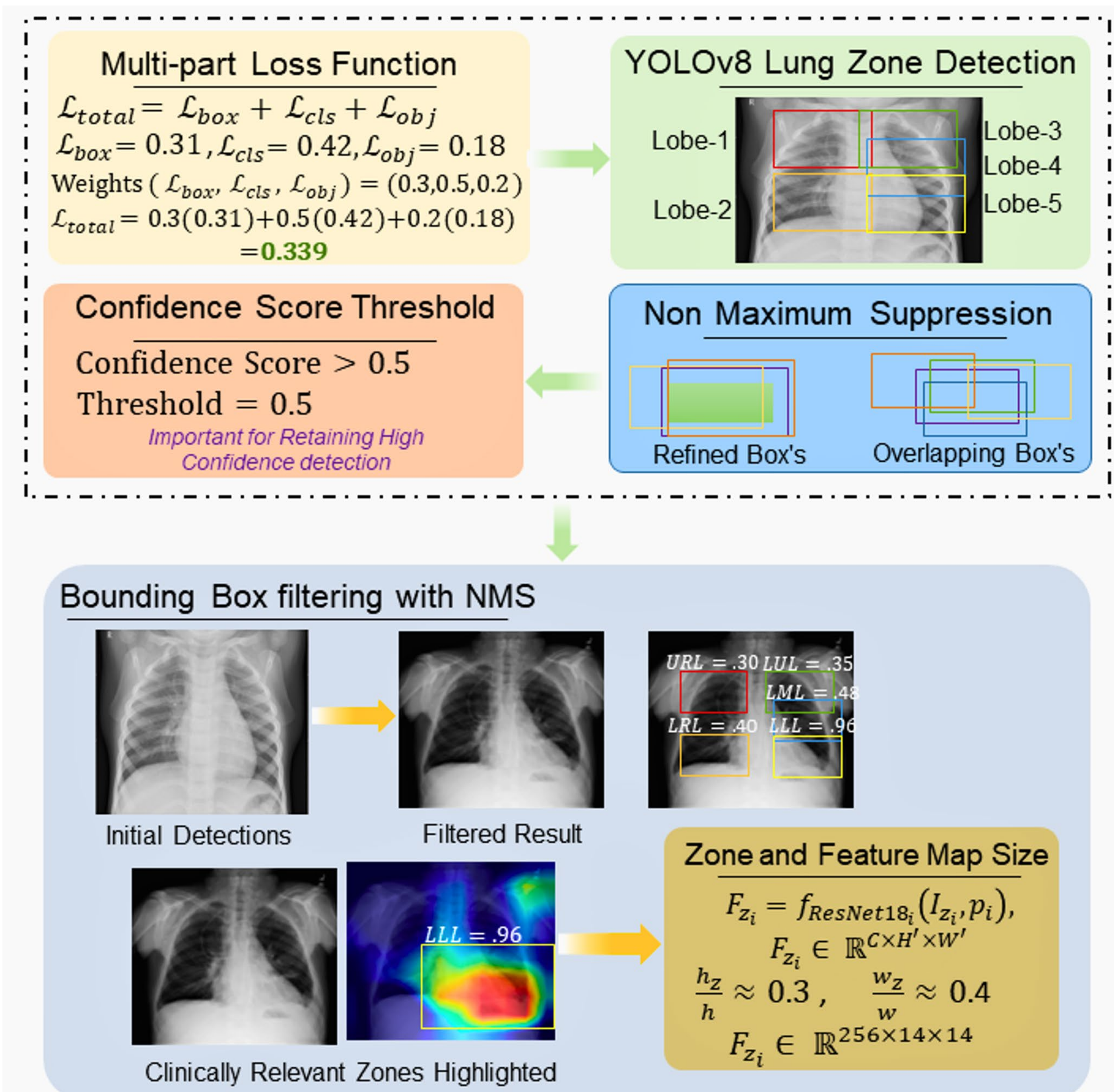


Fig. 2 Visual representation of YOLOv8 based lung zone detection and multi-zone feature extraction

Algorithm Multi-branch feature aggregation

Input: *Zone level feature vectors* $\boldsymbol{F}_{z_1}^{cot}, \boldsymbol{F}_{z_2}^{cot}, \boldsymbol{F}_{z_3}^{cot}, \boldsymbol{F}_{z_4}^{cot}, \boldsymbol{F}_{z_5}^{cot}$
Parameters: *learnable vector* $\boldsymbol{w}$*; fusion block* $\boldsymbol{g}(\cdot)$*; classifier* $(\boldsymbol{W}, \boldsymbol{b})$
Output: *Predicted multi-label vector* $\boldsymbol{\mathcal{Y}} \in [\boldsymbol{0}, \boldsymbol{1}]^{N_{attribute}}$
Step 1: Intra-Zone CoT
For each zone $\boldsymbol{i} \in \{\boldsymbol{1}, \dots, \boldsymbol{5}\}$:
1.1 Compute unnormalized score: $\boldsymbol{s_i} \leftarrow \boldsymbol{exp}(\boldsymbol{w}^{\intercal}\boldsymbol{F_i})$
1.2 Normalize across zones (softmax):

$$\boldsymbol{\alpha_i} \leftarrow \frac{\boldsymbol{s_i}}{\sum_{j=1}^{5} \boldsymbol{s_j}}$$

Step 2: Intra-Zone reweighting
For each zone $\boldsymbol{i}$:

$$\widetilde{\boldsymbol{F}_i} \leftarrow \boldsymbol{\alpha_i}.\boldsymbol{F_i}$$

Step 3: Inter-zone CoT
3.1 $\boldsymbol{G_1} \leftarrow \boldsymbol{g}(\widetilde{\boldsymbol{F}}_{z_1}, \widetilde{\boldsymbol{F}}_{z_2})$
3.2 $\boldsymbol{G_2} \leftarrow \boldsymbol{g}(\boldsymbol{G_1}, \widetilde{\boldsymbol{F}}_{z_3})$
3.3 $\boldsymbol{G_3} \leftarrow \boldsymbol{g}(\boldsymbol{G_2}, \widetilde{\boldsymbol{F}}_{z_4})$
3.4 $\boldsymbol{G_4} \leftarrow \boldsymbol{g}(\boldsymbol{G_3}, \widetilde{\boldsymbol{F}}_{z_5})$
Step 4: Final aggregation:

$$\boldsymbol{F}_{\text{final}} \leftarrow \text{Concat}(\boldsymbol{G_4}, \widetilde{\boldsymbol{F}}_{z_1}, \widetilde{\boldsymbol{F}}_{z_2}, \widetilde{\boldsymbol{F}}_{z_3}, \widetilde{\boldsymbol{F}}_{z_4}, \widetilde{\boldsymbol{F}}_{z_5})$$

Step 5: Prediction:

$$\boldsymbol{\mathcal{Y}} = \boldsymbol{\sigma}(\boldsymbol{W}\boldsymbol{F}_{\text{final}} + \boldsymbol{b})$$

Step 6: *Return* $\boldsymbol{\mathcal{Y}}$

$$F_{final} = Concat\left(G_4,\ \widetilde{F}_{z_1}, \widetilde{F}_{z_2}, \widetilde{F}_{z_3}, \widetilde{F}_{z_4},\ \widetilde{F}_{z_5}\right)\ \in\ \mathbb{R}^{6\times\, d} \tag{12}$$

Concatenation is performed to retain the global reasoning context and integrity of each lung zone feature vector in a fixed positional sequence. This also maintains spatial features and prevents early inter-zone feature mixing before final prediction. The unified fused vector F_{final} is then passed through a fully connected layer:

$$\mathcal{Y} = \sigma\left(WF_{final} + b\right) \tag{13}$$

Where: $W \in\ \mathbb{R}^{5\times\, d}$ is the weight matrix, $b \in\ \mathbb{R}^{5\times\, d}$ is the bias vector, σ represents the sigmoid activation function, and $\mathcal{Y}$ is the final output labels, including pneumonia type and structured report attributes.

For training, the total classification loss is optimized and defined as:

$$\mathcal{L}_{total} = -\frac{1}{N}\sum{}_{n=1}^{N} CE(\mathcal{Y}^{(n)}, \widehat{\mathcal{Y}}^{(n)}) \tag{14}$$

Where, $\mathcal{L}_{total}$ is the total loss function, $\mathcal{Y}^{(n)}$ is the ground truth label for sample n, and $\widehat{\mathcal{Y}}^{(n)}$ is the model's prediction from the fused features representation. Overall, the comprehensive visual representation of zone detection is presented in Fig. 2 that highlights the model ability how to classify the zones, bounding boxes regression, and multi-zone feature learning stages.

The above defined feature fusion stage ensures that each zone's feature is retained before the final prediction. By combining them step-by-step, the model learns how zones relate, making it easier to detect signs like bilateral opacities or disease spread across zones. The feature fusion is implemented using a simple algorithm:

3.5 Model Interpretability and Feature Attribution

This step provides visual feedback for zone-specific feature contributions. It employs the Explainable AI (XAI) technique, Grad-CAM (Gradient-weighted Class Activation Mapping), to highlight the most influential regions in each lung zone image.

Here, the final prediction score for class c is represented as denoted as $\mathcal{Y}_c$. The feature map generated from a CNN branch for the lung zone z_i is defined as:

$$F_{z_i} \in \mathbb{R}^{C\times H' \times W'} \tag{15}$$

where C is the number of channels, and $H' \times W'$ are the spatial dimensions of the feature map. Grad-CAM computes the gradient of $\mathcal{Y}_c$ with respect to the feature map F_{z_i}:

$$\frac{\partial \mathcal{Y}_c}{\partial F_{z_i}} \tag{16}$$

To generate the Grad-CAM heatmap for the zone z_i, first calculate channel-wise importance weights:

$$\alpha_k = \frac{1}{H' \times W'} \sum_{p=1}^{H'} \sum_{q=1}^{W'} \frac{\partial \mathcal{Y}_c}{\partial F_{z_i}^{k,p,q}} \tag{17}$$

Where α_k is the importance weight of the k-th channel, and $F_{z_i}^{k,p,q}$ represents the value at spatial location *(p, q)* in channel k.

The Grad-CAM heatmap $L_{Grad-CAM}^{z_i}$ is computed as:

$$L_{Grad-CAM}^{z_i} = ReLU\left(\sum_{k=1}^{C} \alpha_k . F_{z_i}^{k}\right) \tag{18}$$

Where ReLU ensures that only features with a positive influence on the prediction contribute to the heatmap. Each zone z_i produces its own heatmap $L_{Grad-CAM}^{z_i}$, which is resized to the size of I_{z_i} using bilinear interpolation.

This method ensures that model decisions are transparent to clinicians. By visualizing which parts of each lung zone influenced the prediction, users can confirm whether the model's focus matches known medical patterns. It also helps detect model errors. For example, if Grad-CAM highlights irrelevant areas outside lung zones, it indicates potential overfitting or poor training.

The system applies Grad-CAM on the last convolutional feature map of each zone branch. In addition, the zone attention weights $\boldsymbol{\alpha}_i$ obtained from fusion are highlighted to ensure no features from other zones interfere with the heatmap of a specific zone. Moreover, Grad-CAM is also applied on individual zones to show the importance of each zone. This modular XAI approach respects the structure set up in the earlier steps of the model. The same procedure can be applied during training to monitor feature learning or post-training for result interpretation. It provides a reliable and structured way to explain multi-zone pneumonia classification decisions.

Table 1 Overall pneumonia classification performance

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
ResNet-18	84.7	84.3	84.0	84.1	0.88
ViT	86.7	86.2	85.9	86.0	0.89
CNN+Attention (baseline)	88.8	88.3	88.0	88.1	0.91
Proposed (YOLO+Dual CoT)	94.4	94.9	93.8	94.3	0.97

4 Results

This section presents the evaluation of the multi-stage pneumonia detection and classification framework using chest X-ray images. The results include every stage of the pipeline, starting from lung zone detection to the final pneumonia classification. Each stage is tested step-by-step to measure its contribution to the overall performance. Unlike traditional baseline models (CNN or Transformer), which process the entire chest X-ray as one image. This proposed model introduces a dual Chain-of-Thought reasoning mechanism. It sequentially accumulates local sub-features at the intra-zone level and captures cross-zone dependencies at the inter-zone level. This framework is novel as it combines lung zone detection, intra-zone reasoning, and inter-zone reasoning for accurate analysis and detection of lung infection. The performance of this proposed model is evaluated with quantitative results (tables) and visual representation (performance graphs, cropped zones, and heatmaps). Furthermore, an ablation study is presented to measure the effect of removing specific components from the pipeline. Additional experiments report zone-wise performance and the effect of class imbalance.

4.1 Overall Performance

Table 1 presents the overall pneumonia classification performance across the proposed model and other baseline models. These traditional whole-image processing models, such as ResNet-18 and ViT, achieve accuracies of 86.7% and 84.7%, respectively. A CNN baseline with attention improves accuracy to 88.8%. The proposed model integrates YOLO-zone detection with attention-based dual Chain-of-Thought reasoning, achieving the highest accuracy, results across all metrics. This highlights the value of zone-level reasoning and cross-zone fusion compared with whole-image processing.

4.2 Zone Detection (YOLOv8)

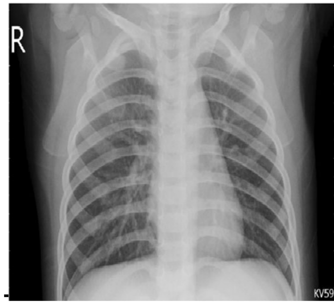
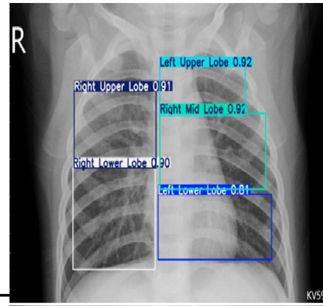
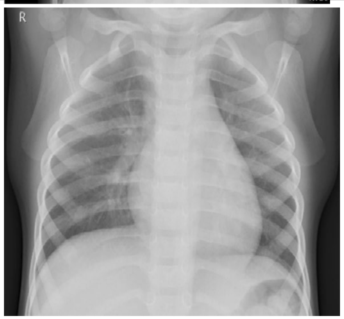
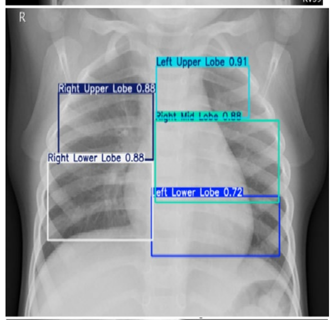
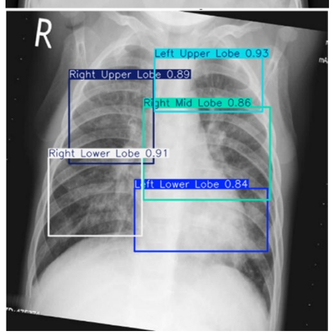
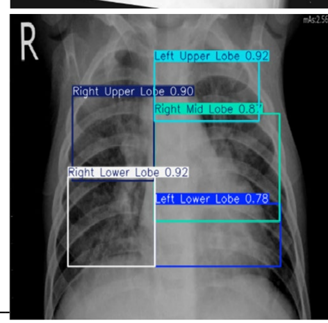
The first stage of the model is to locate each lung region in each chest X-ray. These are the Right Upper Zone (RUZ), Right Mid Zone (RMZ), Right Lower Zone (RLZ), Left Upper Zone (LUZ), and Left Lower Zone (LLZ). Accurate zone detection is important, as each zone is linked to different disorders. Detecting each zone separately helps the model to focus on the diseased area and overlook healthy regions. YOLOv8 is used for strong localization due to its fast inference speed. The model predicts bounding boxes around each zone with minimal overlap and clear boundaries. Consistent detection across all zones ensures clean inputs for the next feature extraction and classification.

Detection performance is shown in Table 2, measured by metrics such as mAP@0.5, precision, and recall. The metric mAP is called as mean Average Precision that is computed at intersection over union (IoU) with a threshold of 0.5. This mAP@0.5 is specifically used to decide whether the predicted bounding box is a correct decision or not. If IoU corresponding to ground truth is greater than or equal to 0.5 the bounding box is considered as correct detection. This mAP at 0.5 IoU is considered as widely adopted threshold in object detection and it give a clear evaluation about accurate localization. The model achieves 96% of average mAP@0.5, 95% of average precision, of 94% average recall. Among all zones, the RLL zone achieves the highest 97% mAP@0.5 and 96% precision and 95% recall, which indicates that the model is effectively identifying this area. The LUL zone has

Table 2 Detection performance for lung zones

Zone	mAP@0.5	Precision	Recall
Right Upper Zone	0.962	95%	94%
Right Middle Zone	0.951	94%	93%
Right Lower Zone	0.973	96%	95%
Left Upper Zone	0.949	94%	93%
Left Lower Zone	0.965	95%	94%
Avg	0.960	95%	94%

Table 3 Qualitative analysis of YOLOv8 on each lung zone

Original Chest X-ray	YOLOv8 Detected Zones	Anatomical Coverage	Inter-Zone Separation	Localization Stability
		Upper, middle, and lower lung zones are correctly covered according to anatomical landmarks	Clear boundaries between zones with no visible overlap	Stable localization under normal imaging conditions
		Complete lung field partitioning achieved with correct vertical ordering of zones	Zones remain spatially independent, ensuring clean cropping	Robust to mild geometric variations
		Detected regions align with expected lung anatomy despite uneven exposure	Minimal boundary interference observed	Localization preserved under mild rotation
		Upper, middle, and lower zones correspond to standard radiological reference areas	Non-overlapping zone regions maintained	Consistent spatial placement across images

the lowest 95% mAP@0.5, but this detection rate is still high, highlighting only a small variation in performance across zones. These results show that YOLOv8 provides strong, consistent, and accurate localization of all lung zones, which act as a strong base for the next steps in the pipeline.

Table 3 shows lung zone detection results from YOLOv8. Each zone is marked with a bounding box that matches the correct anatomy. Upper, middle, and lower zones have clear boundaries with minimal overlap. The bounding box's location is clearly aligned with anatomical coverage, inter-zone separation is clearly defined even under mild rotation or uneven exposure. This ensures each cropped zone contains the right anatomy without including background regions.

4.3 Zone-wise Performance

The classification performance of the proposed model across each lung zone in chest X-ray is reported in Table 4. The model maintains consistently high performance in all five zones, which provides reliability in analyzing lung zones separately. Results are more accurate in the right lung as the Right Upper Zone achieves 91.2% accuracy, and 90.5% accuracy in the Right Lower Zone. While the model achieves 87.8% accuracy at the Left Upper Zone, due to overlapping with the heart shadow.

These results show the model adapts to zone complexity while improving efficiency. This zone-level evaluation also highlights clinical interpretability. Radiologists not only require a "pneumonia yes/no" decision but also zone-specific findings that show where the abnormality is located. This framework provides zone-specific outputs that make it more useful in a clinical setting than black-box models.

4.4 Ablation Study

To evaluate the impact of different components in the framework, an ablation study is presented in Table 5. The baseline CNN model processes entire CXR images and performs worst, with the lowest accuracy of 83.2% and 0.874 AUC, showing the weakness of global-only models. Moreover, adding YOLO zone detection improves accuracy to 85.4%, highlighting that localized lung regions are helpful in classification. Furthermore, integrating intra-zone CoT improves accuracy to 87.6% and F1-score to 86.9%, which demonstrates its ability to capture subtle features within zones. Next, Inter-zone CoT increases accuracy to 88.2%, indicating that cross-zone reasoning helps to detect bilateral or multi-zone involvement. Attention-based fusion also improves results but performs slightly weaker than CoT-based reasoning, proving attention mechanism alone is less effective. The entire model, with YOLOv8, dual CoT, and attention, achieves the best performance with 94.4% accuracy and 0.971 AUC. Each module improves overall performance, but integrating all provides the most balanced system. These results validate the novelty of the proposed model and highlight its advantage over whole-image classification black-box models.

4.5 Effect of Class Imbalance Handling

This experiment is conducted to evaluate the impact data imbalanced on model's robustness. The dataset Chest X-Ray Images (Pneumonia) dataset [41] exhibits inter-class data imbalance where 4273 Performance samples are included which is higher than are 1590 normal samples. To handle this class imbalancing weighted cross-entropy and focal loss function are employed during model optimization stage that reduce the impact of majority class. then zone aware attention and CoT reasoning help to keep model focus on zone specific pneumonia regions that prevent the chances of model to train against majority class features. This experiment also highlights the proposed model behavior as compare to different loss function that can be utilized to handle the class imbalance results are presented in Table 6.

These results show that Standard cross-entropy is less effective as it shows 80.1% recall on minority classes, while weighted CE and focal loss show improvements. However, the proposed model with dual CoT reasoning

Table 4 Zone-wise performance of proposed model

Zone	Accuracy (%)	Sensitivity (%)	Specificity (%)
Right Upper Zone	91.2	89.5	92.0
Right Middle Zone	88.7	87.1	89.4
Right Lower Zone	90.5	88.9	91.2
Left Upper Zone	87.8	85.6	88.9
Left Lower Zone	89.3	87.5	90.1

Table 5 Effect of removing and adding key modules on the overall performance

Model Variant	Accuracy (%)	F1-score (%)	AUC
CNN (whole CXR) only	83.2	81.8	0.874
+ YOLO zone detection only	85.4	84.2	0.891
+ Intra-zone CoT only	87.6	86.9	0.912
+ Inter-zone CoT only	88.2	87.5	0.919
+ Attention (no CoT)	89.9	87.2	0.914
Full Model (YOLO + Dual CoT + Attention)	**94.4**	**94.3**	**0.971**

Table 6 Effect of class imbalance handling

Loss Function	Accuracy (%)	F1-score (%)	Minority-class Recall (%)
Cross-Entropy (baseline)	86.2	86.0	80.1
Weighted CE	88.1	87.8	83.5
Focal Loss	87.4	87.0	82.2
Proposed (Weighted + CoT)	**94.4**	**94.3**	**87.7**

Table 7 Comparison with existing approaches

References	Year	Method	Dataset	Zone Level Analysis	Accuracy (%)	F1-score ()	AUC
[47]	2022	Attention + Modified DenseNet	Chest X-Ray Images (Pneumonia)	Global feature analysis only	93	94.5	–
[48]	2022	DenseNet169 + MobileNetV2 + ViT	Chest X-Ray Images (Pneumonia)	Global feature analysis only	94	93	–
[49]	2023	VGG16 with dense CNN	Chest X-Ray Images (Pneumonia)	Global feature analysis only	92	93	0.97
[50]	2023	CNN Ensemble Model	Chest X-Ray Images (Pneumonia)	Global feature analysis only	84.4	88.56	0.93
[51]	2024	EfficientNetB0 + DenseNet121 with Attention	Chest X-Ray Images (Pneumonia)	Limited region-level attention but on whole image	95.1	96.0	0.95
[52]	2025	ResNet50 + interpretability (LRP, CAMs, adversarial training, spatial attention)	Chest X-ray Images (Pneumonia)	Coarse region highlighting in whole image	91	–	0.93
Proposed	2025	YOLO + Dual CoT + Attention	Chest X-Ray Images (Pneumonia)	Fine-grained zone-aware localization	**94.4**	**94.3**	**0.97**

Table 8 Grad-CAM evaluation

Zone	% of CAM correctly aligned with annotated lesion
Right Upper Zone	88
Right Middle Zone	85
Right Lower Zone	87
Left Upper Zone	82
Left Lower Zone	84
Average	**85.2**

and attention mechanism achieves the best result, 94.4% accuracy, 94.3% F1-score, and 87.7% minority-class recall. These results show that zone-aware reasoning prevents minority features from being suppressed. The framework provides balanced performance across all pneumonia classes.

4.6 Comparison with Existing Approaches

To highlight the ability of this proposed model a comparison is presented in Table 7 with other state-of-the-art art widely used existing approaches. This comparison is based on CNNs, multimodal (CNN-ViT), implementation of XAI interpretability techniques, and attention based modified techniques. Such as DenseNet, specifically trained in CheXNet style, achieves 93.1% accuracy and 94.5% F1-Score. A deep learning-based hybrid model is applied that integrate multiple models DenseNet169, Mo-bileNetV2 and Vision Transformer to improves results to 94.3% accuracy and 93% F1-Score, which highlights the advantage of using Transformer-based architectures in optimized feature learning. Although these techniques demonstrate high performance but these hybrid model integrates local and global features and they treat chest x-ray as a single global entity and trained on whole X-ray images not on specific zones that focus on clinically relevant regions. The proposed model with YOLOv8 and dual Chain-of-Thought reasoning achieves the best outcome, 94.4% accuracy, 94.3% F1-Score, and 0.97 AUC. These results demonstrate that zone-aware detection and sequential reasoning capture clinically relevant patterns that are more effective and robust than whole-image classifiers.

4.7 Explainability

This experiment is conducted for the evaluation of zone-level explainability through Grad-CAM. The performance result is presented in Table 8, which illustrates how correctly this Grad-CAM aligns with the annotated lesion. On average, the model aligns 85.2% between highlighted regions and annotated lesions. Right Upper Zone is 88% correct, and Right Lower Zone shows 87% aligned with the annotated lesion. Left Upper zone is 82% correct, which is the lowest due to heart shadow interference. These results confirm that the heatmaps match with expert annotations and focus on lesion areas across most zones. The ability to highlight true lesion areas supports clinical trust and makes prediction transparent and meaningful for radiologists.

The attention heatmaps generated by proposed model are illustrated in Fig. 3 that reveals the most clinically relevant zones of the image on which the hybrid model focuses while making decisions for better interpretation. The visualizations demonstrate how features are localized, and global features contribute to classification.

5 Conclusion

Pneumonia is a major health problem all around the world that can badly affect the people of all ages and its timely and accurate identification is important for its effective treatment. Automated analysis with precise and confident detection using AI-based systems demonstrate a strong potential to help radiologist in decision making. Many advance approaches are developed that rely on whole x-ray image explicitly overlook zone level detail. To overcome this, an automated framework is introduced in this study for pneumonia detection and reporting using chest X-rays. The system integrates lung zone detection with attention-based dual CoT feature extraction and classification. It identifies pneumonia type and reports zone-level findings such as opacities and ground-glass changes. By focusing on lung regions, the model reduces background noise and improves accuracy and interpretability. Experimental analysis performs on the Chest X-Ray Images (Pneumonia) dataset demonstrated an average 94% accuracy and accurate detection of early-stage pneumonia. Importantly, the system also generates a heatmap of clinically relevant features aligned with clinical practice, supporting trust for radiologists. These

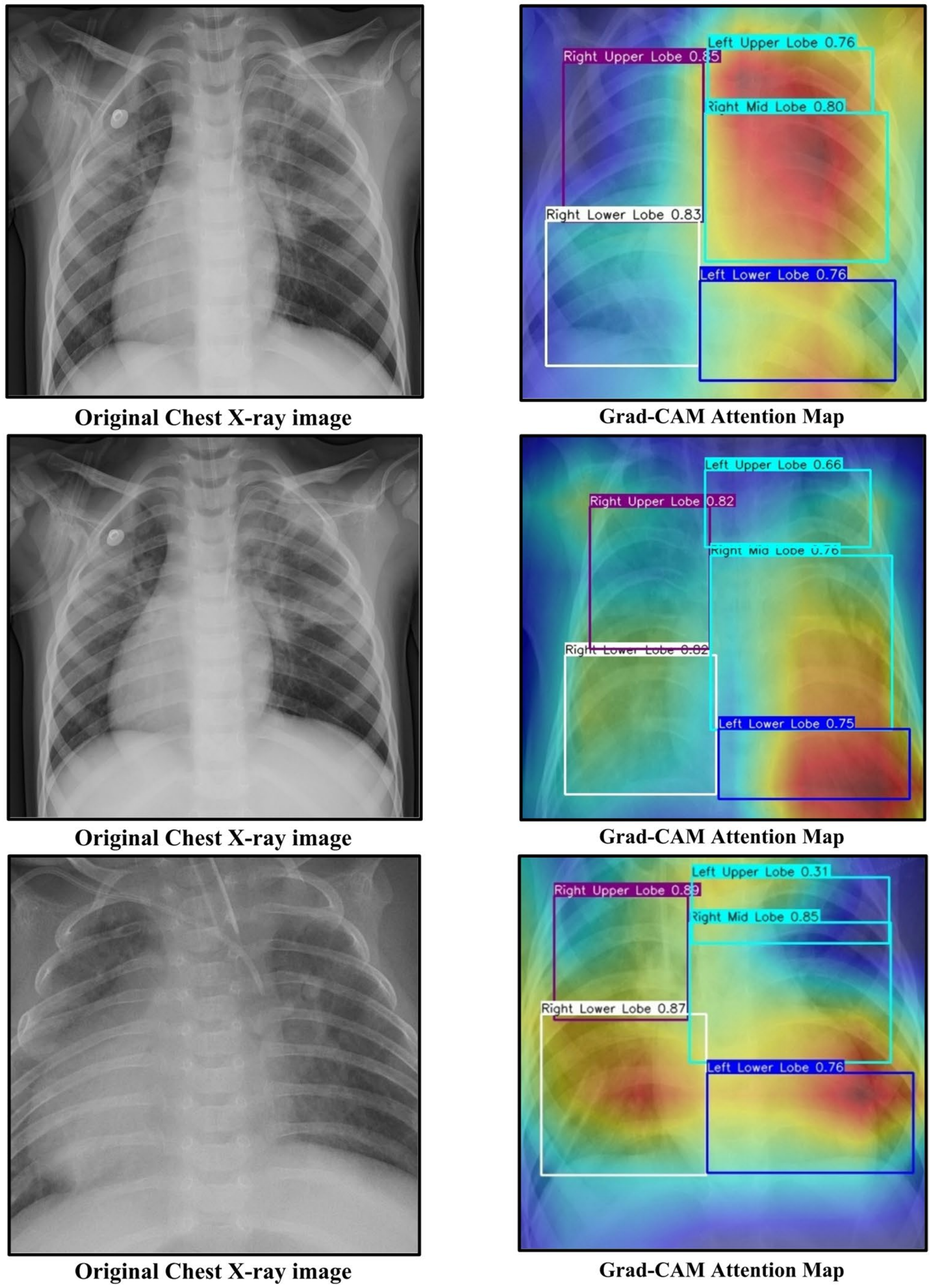


Fig. 3 Zone-level Attention Maps generated by Grad-CAM

contributions highlight that zone-aware deep learning can improve diagnostic consistency and reduce workload. Future work will include validation of the system on multi-center datasets and integration into clinical platforms.

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Data availability The Chest X-Ray Images (Pneumonia) datasets analyzed during this study are publicly available in the Kaggle dataset repository and can be accessed at [https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia](https:/www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia) , we also test this proposed model on [https://www.kaggle.com/datasets/artyomkolas/3-kinds-of-pneumonia](https:/www.kaggle.com/datasets/artyomkolas/3-kinds-of-pneumonia) . Additionally, the code for this study is provided in code link [https://github.com/jhshah101/Zone-Aware-Chest-X-Rays-CoT/tree/main](https:/github.com/jhshah101/Zone-Aware-Chest-X-Rays-CoT/tree/main).

Declarations

Competing interests The authors declare no competing interests.

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